

LECTURES IN CRITICAL THINKING AND PROBLEM SOLVING

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STUDY/SELF-ASSESSMENT QUESTIONS



Lecture Series By:

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CRITICAL THINKING AND PROBLEM-SOLVING STUDY QUESTIONS

Note: Most of the questions listed below are from past examination papers for Critical thinking and problem-solving. The purpose of these questions is to guide and help learners to be conversant with examination setting format and at the same time to enable them to prepare well for the examination. The list is however not exhaustive and further reading beyond the list is recommended.

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PHILOSOPHY

1. Define philosophy and show its relevance in human life.
2. What are the uses of philosophy?
3. By way of examples, show the relationship between Philosophy and Critical thinking.
4. Discuss *any two* of the following branches (areas) of Philosophy
 - a) Logic
 - b) Epistemology
 - c) Metaphysics
 - d) Axiology
 - e) Ethics
 - f) Aesthetics
5. To what extent is philosophy a First order discipline?
6. Examine the view that logic is more of a first-order principle to philosophy than its branch.
7. “Philosophy is about the nature of, and essence of reality itself” Discuss.
8. “Everyone who is capable of thinking is a philosopher so there is no need of studying philosophy” Discuss.

9. “All that philosophers do is make life more difficult than it is” Do you agree with this statement? Give reasons.
10. What is the difference between a Philosopher and an African sage?

HISTORY OF CRITICAL THINKING

11. Discuss the Socratic dictum that “an unexamined life is not worth living”
12. Give a brief history of Critical thinking.
13. Of the following three philosophers; Socrates, Rene Descartes, and Bertrand Russell, whom would you consider the *Father of Critical thinking and problem-solving*? Give reasons for your answer.
14. What is the relevance of understanding the evolution of critical thought to the study of critical thinking?
15. To what extent does the study of history of Critical thinking help in the understanding of the development of Critical thought?
16. Define Critical thinking and show its relevance in an individual’s life

THE SOCRATIC QUESTIONING METHOD

17. Explain why the Socratic Method best exemplifies Critical thinking and problem solving
18. Highlight some of the positive lessons a student of critical thinking can learn from Socrates.
19. “An unexamined life is not worth living” To what extent is this Socratic Maxim relevant as a guide to a critical thinker?

DEFINITIONS, NATURE, AND CHARACTERISTICS OF CRITICAL THINKING

20. “Kenya would be a better country to live if its leaders were guided by reason and not emotion and self-gratification” By way of examples discuss this statement in the context of Critical thinking tools of analysis and tools of evaluation.
21. What are the main characteristics of Critical thinking?

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22. Discuss the extent to which the study of Critical thinking and problem-solving has helped in enhancing performance in your studies/work.
23. “The incessant bickering between Counties and National Government is not because of scarce resources, but the inability of our leaders to think critically” Discuss.
24. What are the essential aspects of critical thinking?
25. “Critical thinking and problem solving should be the basic induction training program to all elected Members of Parliament, Senators, Governors, Women Representatives, and MCAs before they occupy office”. Giving reasons, state why you agree or disagree.
26. Examine the view that Critical thinking and problem solving is essential for good life.
27. By way of specific examples critically examine the dimensional categories (tools of analysis) of thinking
28. Using at least **three** Critical thinking Tools of Analysis explain the relevance of Moreno O’Campo’s argument against the “O’Campo Six” that they bear the greatest responsibility in Kenya’s 2007 post-Election Violence.
29. With reference to the ongoing review of the education system in Kenya, examine the view that Critical thinking and problem solving ought to inform the new curriculum.
30. Examine critical thinking tools of evaluation showing their role in checking the quality of proper thinking (reasoning)
31. Discuss critical thinking tools of analysis showing the intellectual standards (tools of evaluation) relevant to their application.
32. “The study of Critical thinking and problem solving should be made mandatory in all institutions for Higher Learning” Do you agree or disagree? Give reasons.
33. By way of examples discuss the relevance of any four tools of transformation (Valuable intellectual traits).
34. Show how Critical thinking and problem solving constitutes the core meaning of philosophy.
35. Critically examine the seven traits of mind and then show their relevance in enhancing roles and responsibilities for self-management.
36. Discuss the characteristics of Critical thinking and then show their relationship to the universal intellectual standards.

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37. In view of the escalating state of insecurity in Kenya and the rising cost of living coupled with the growing cases of corruption being cited, how do you assess and rate Kenyan leaders in embracing criticality and creativity in their decision making and problem solving strategies?
38. Illustrate the ways and means the following elements are involved in Critical thinking and problem solving;
- a) Reasoning skills
 - b) Collaborative learning
 - c) Self-esteem
 - d) Curiosity
39. Explain any four intellectual traits/tools of transformation and justify why they should be enhanced in Kenya's political life.
40. Give examples to illustrate the main elements of critical thinking. How to reasoning skills, self-esteem and interactive learning fit in?
41. Using examples to illustrate your answer discuss the following:
- a) Common core thinking skills
 - b) The essential components of critical thinking
 - c) The dispositions to be cultivated by a critical thinker
42. Examine the major intellectual traits and then show their relevance in determining the character of a critical and creative thinker.
43. "There is no point in studying Critical thinking and problem solving since all normal human beings think." Give reasons for your answer.
44. Discuss the relevance of Critical thinking tools of transformation as essential traits to a critical and creative thinker.
45. "Critical thinking and problem solving enhances the quality of an individual's life" Discuss

EDUCATION, IDENTITY AND SELF EXAMINATION

46. "True education is that which leads to the holistic growth of the human person". Discuss this statement with reference to traits, attitudes, and dispositions of education for human development.

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47. Education that transforms is the only education of which it can be truly said that it makes good citizens. Discuss
48. With reference to education for human development explain the meaning of two of the following:
- a) Essential Skills for thinking
 - b) Dispositions for Critical Thought
 - c) Traits of a Critical Mind
 - d) Self-examination
49. In reference to Russell's work on education, examine the view that Education in the critical faculty is the only education of which it can be truly said that it makes good citizens.
50. Discuss the role of critical thinking in education for self-actualisation
51. Who am I? Discuss
52. What is the role of critical thinking in an individual's life?
53. "I am who I am" to what extent does this statement define the self of an individual?
54. "The only way to eradicate poverty is to empower the economically deprived through education." Do you agree Give reasons?
55. "I am because we are and since we are therefore, I am" To what extent does this statement negate the principle of self-identity?
56. "Of the three persons in me, namely, that whom people think I am, that whom I think I am and the unknown me, there lies the really me." Espouse on this statement.

THINKING FOR ONESELF: Rene Descartes

Cogito ergo sum (I think, therefore I am)

57. Of what relevance is the Cartesian Methodic Doubt to the study of Critical thinking and problem-solving?
58. Examine the view that Cartesian Methodic Doubt is a precursor to Critical thinking and problem-solving.
59. Examine the view that Rene Descartes' Mathodic Doubt exemplifies Critical thinking and problem-solving.

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60. Critically examine the relevance of René Descartes dreams to his call as a philosopher and/or critical thinker.
61. What are the main philosophical foundations that informed the Cartesian methodic Doubt?
62. Critically examine the relevance of the Cartesian methodic Doubt to the process of self-knowledge.
63. By way of examples discuss the relationship between the Cartesian Methodic Doubt and the subject of Critical thinking and problem-solving.
64. By way of examples show the relevance of Cartesian Methodic Doubt to the process of self-examination and the knowledge of the self.

APPEARANCE AND REALITY - Bertrand Russell

65. Using relevant examples demonstrate the extent to which conmen, politicians, advertisers, street/rogue preachers, and propagandists take advantage of appearance as reality. “
66. To have a better understanding of the world in which we live, we must strive to distinguish between Appearance and Reality” Discuss.
67. “Senses are deceptive and it is only by use of reason that we can claim knowledge” Discuss this statement in the light of Bertrand Russell’s Appearance and Reality.
68. a) By way of examples demonstrate the problem of Appearance and Reality
b) Show why a critical and creative thinker ought to distinguish between Appearance and Reality.
69. To what extent do conmen, advertisers, politicians and street preachers take advantage of appearance and reality?
70. Discuss the relevance of Appearance and Reality in our lives.
71. How can one apply Russell’s thought on “Appearance and Reality” to everyday life in Kenya today?
72. Discuss how our understanding of the universe is enhanced by distinguishing Appearance from Reality.

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ON LIBERTY OF THOUGHT AND INDIVIDUALITY – John Stuart Mill

73. “Freedom is only relevant insofar as it does not infringe on the liberties of others”. Discuss this statement with reference to J.S. Mills’ essay on Individuality.
74. “Individuality is a necessary condition to social progress in human society”. Discuss.
75. Give a brief history of John Stuart Mill’s life and then discuss his views on human liberty and freedom as expounded in *On Liberty*
76. “The attainment of autonomy is the ultimate goal in human development” Discuss this in light to J.S. Mills essay on essay on Individuality and Liberty of Thought
77. Critically examine John Stuart Mill’s view that attainment of autonomy is the prima facie principle for human existence.
78. What are the basic elements of individuality?
79. How can individuality enhance the wellbeing of society?
80. Individuality is crucial to Critical thinking. Is this justifiable given man’s sociality?
81. Show the connection between individuality and freedom in the context of Critical thinking and problem solving.
82. In your view, what would you consider a fair relationship between social authority and individual autonomy in social life?
83. What do you consider to be the place of an individual in the state?
84. a) What are slogans?
b) Can slogans contribute positively to human well-being?
85. Autonomy is the ultimate goal of human development. Discuss this in the light of J.S. Mill’s essay on Individuality.

LOGIC

86. Using your own examples demonstrate that the distinction between Inductive and Deductive reasoning is based on the degree of the evidential strength between premises and conclusions of arguments.
87. Define argument and then distinguish between the following types of arguments:
88. Define logic and then show its relevance in our everyday day life.
89. Distinguish between deductive and inductive reasoning.

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90. Indicate the premises and conclusions of the arguments contained in the following passage:

With regard to good and evil, these terms indicate nothing positive in things considered in themselves, nor are they anything else than modes of thought, or notions which are formed from the comparison of one thing with another. For one and the same thing may at the same time be both good and evil or indifferent. Music for example, is good to a melancholy person, bad to one mourning, while to a deaf man it is neither good nor bad....

91. What is the meaning of:

- a) an inference
- b) a deductive argument
- c) an Inductive argument

92. Distinguish between Truth and Validity

93. Show the relevance of deductive and inductive reasoning to the study of Critical thinking and problem-solving.

94. **Recognizing the Form of Simple Arguments**

Here are some short arguments in which the writer presents a conclusion (which is in bold) and provides some reasons for that conclusion.

Indicate beside each argument whether it is an example of deductive or inductive reasoning (you can use the letters D and I). If you are not sure, use a question mark.

Note that this exercise is not asking you whether you agree with the argument or not or whether the argument is a good one or not. It is asking you only to indicate the form of reasoning used, inductive or deductive. Remember the key test here: Does the argument rely upon an appeal to a general principle or upon assembled data.

1. Things equal to the same thing are equal to each other. Therefore if A equals B and if B equals C, **then A must equal C.**

2. The doctrine of free speech is the most important element of our liberal democracy. **Therefore, this student newsletter must be free to print opinions offensive to many people.**

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3. Six out of ten test samples of the water in that lake, collected and analyzed by university researchers last week, revealed unsatisfactorily high levels of serious contamination. **We must investigate this problem further and post warning signs on the beach immediately.**

4. All human beings have the right to die with dignity when they wish. **Therefore this terminally ill patient has the right to an assisted suicide.**

5. In this essay the writer frequently uses words like "perhaps," "maybe," and "alternatively." **This feature of the style creates doubts in the mind of the reader about the writer's confidence in his analysis.**

6. Giving minority groups the right to political self-determination is fundamental to liberty. Therefore, if a majority of Rift valley people vote for independence from Kenya, **they must be allowed to separate.**

7. All people in a free society must be treated equally under the law. **Homosexual citizens in our society must therefore be granted full legal spousal benefits, equivalent to those of heterosexual Kenyans.**

8. Model X gets better mileage, costs less to purchase and to maintain, and has a better all-around rating in the Consumer Reports than Model Y. **Therefore, it makes more sense for me to purchase Model X rather than Model Y.**

9. Hamlet keeps wondering about why he is not carrying out the murder. He frequently gets upset with himself for delaying, and yet he still seems unable to carry it out. **Clearly, there is something internal preventing him from murdering his uncle.**

10. To what extent do Critical thinking and problem solving improve the power of Rhetoric in an individual?

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FALLACIES

95. "Fallacious reasoning is repugnant to correct thinking" illustrate this statement using four examples two from each of the two general categories of informal fallacies.
96. (a) Distinguish between fallacies of relevance and ambiguity
(b) By way of examples discuss three fallacies of relevance and two fallacies of ambiguity.
97. Using examples, three each from both fallacies of relevance and ambiguity, demonstrate that fallacious thinking is repugnant to good thinking.
98. Discuss any **three** fallacies showing how they can be avoided. In each case furnish relevant examples.
99. Give **three** examples of fallacies of relevance and in each case, show how one is misled by the argument.
100. Fallacious reasoning is offensive to correct thinking. Illustrate this using four examples, two from each of the two general categories of informal fallacies.
101. Give **three** examples of fallacies and show why they are instances of incorrect reasoning.
102. By way of examples, distinguish between fallacies of relevance and fallacies of ambiguity.
103. To what extent can the study of Critical thinking and problem-solving help one in avoiding fallacious reasoning?
104. Identify the type of fallacy committed in each of the following statements:
 - a) "I'd like to respond to that, but taking into account your background, education, and intelligence, I am quite sure that you would not be able to understand."
 - b) "I have observed that those who disagree with me on the next point tend to be unsophisticated, and those who quickly recognize the validity of the point to be more educated. The point is...."
 - c) "As you mature emotionally (or mentally, or spiritually), you will grow out of your present way of thinking, and you will eventually come around to my point of view."

- d) "You said 'feel' instead of 'think'. If you are feeling instead of thinking, I won't be able to convince you with reason."
- e) "Have I ever brought up the Kshs. 2,000 you owe me? Never! Have I ever embarrassed you or made you feel bad over it? Have I ever told you how much I need that money? No, I never have."
- f) "I agree with everything you said except the conclusion. It doesn't make any sense to me, and I cannot accept it. I am trying, but your brain must work much differently than mine."
- g) "Your problem is that you are thinking in a linear versus configurational framework."
- h) "In view of Kenyatta University budget deficit, staff uprisings, and students' politics, we need to consider that, notwithstanding the mitigating circumstances; this University has got to get back on its feet. Don't you agree?"
- i) "Take this example: suppose you were a person who was incredibly stupid but was trying to come off as intelligent. What would the proper response be if you were me?"

DECISION MAKING AND PROBLEM SOLVING

- 105. a) Define a problem
- b) Critically examine the 6 (six) phases of decision making, showing how they ease the process of problem solving.
- 106. "Being an excellent decision-maker and problem solver is not separable from being a good thinker." Discuss.
- 107. "A good decision maker and problem solver is one that is appropriately moved by reason". Discuss.

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108. You have been appointed the CEO of a leading parastatal whose performance is declining, what decision making and problem solving strategies will you employ in order to salvage the institution?
109. Examine the view that Decision as Intuition negates the principle of rationality.
110. Discuss any **two** of the following types of decision making and show the advantages and disadvantages of each one of them:
- a) Decision as Intuition
 - b) Decision as Calculation
 - c) Decision as Coherence
111. a) What is Emotional Coherence?
- b) Examine the view that Intuition and Emotion can be used in making good decisions
112. In view of the fire and hunger tragedies that have engulfed Kenya in recent times and the escalating cases of corruption being cited, how would you assess and rate our leaders in embracing criticality and creativity in their decision-making and problem-solving strategies?
113. “A promiscuous man has a sexual affair with a Nanny (his house girl) and he tells her – ‘you are so good’. Then the Nanny responds back – “yes thank you, in fact even the houseboy says am better than madam your wife”. How should this man proceed in making a decision that would sort out the mess in his household?
114. Assume that you are a girl aged 14 years and a victim of sexual assault by your own father. If you told your mother who is a housewife the incident is likely to spark off hatred and eventual divorce between your parents. If you keep it to yourself you will continuously be traumatized by this grisly and heinous act. If your mum gets to know of the act via other sources or means she will never forgive you. And if the matter leaks to the wider community your father, the sole bread winner for the family will be jailed or face mob justice that may end his life, leaving you and your siblings helpless and out of school. How would you proceed in making your decision such that it does not hurt any of the parties involved but at the same time be a lesson to the offender?

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BARRIERS TO CRITICAL THINKING AND PROBLEM SOLVING

115. Discuss any *two* of the following Baconian idols and then show how they impede clear thinking,
- a) idol of the tribe
 - b) idol of the market place
 - c) idol of the theatre
 - d) idol of the cave
116. Discuss the extent to which paternalism, religious dogmas, peer influence, and authority constitute obstacles to Critical thinking and problem-solving.
117. Examine the view that a rigid school environment, authoritarianism, peer and social influence, indoctrination, paternalism, and religious fanaticism are impediments to Critical thinking and problem solving.
118. To what extent does the dictatorial political systems affect and block the free use of reason in deliberations?
119. In your view what are the other possible obstacles to clear and creative thinking?
120. To what extent is paternalism, traditions, religion, peer influence, and school environment an impediment to Critical thinking and problem-solving?

STUDY PUZZLES & INTRIGUING QUESTIONS

121. A man is walking to Kisumu town. He comes to a fork in the road, with the two branches leading in two different directions. He knows that one of them goes to Kisumu, but he doesn't know which one. He also knows that in the house right beside the fork in the road there are two brothers, identical twins, both of whom know the road to Kisumu. He knows that one brother always lies and the other always tells the truth, but he cannot tell them apart. What single question can he ask to whoever answers his knock on the door which will indicate to him the correct road to Kisumu?
122. Tom survives a shipwreck in deep waters by hanging on a lifesaver balloon. As the balloon shifts a shore of an island, Tom soberly identifies the island two kilometers away the coastline. The island is inhabited by two distinct tribes, a fact Tom is very much aware of. One of the tribes is called the Nobles while the other the Savages. These two tribes have

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very distinct characters; The Nobles always tell the truth while the Savages always tell lies. Tom is aware of these characteristics and he would like to identify himself with the nobles for positive assistance. As the balloon shifts, Tom sees a man at the shore and he calls out to him: “Are you a noble or a Savage? But when the man answers back, a wave breaks at the shore and Tom fails to hear the answer. The balloon shifts further on and Tom spots another man and he asks of him the same question: “Are you a noble or a savage?” The second man points to the first man and he says: “He said he is a noble” and then adds, “I am a noble.” The balloon is pushed further south by the wind and Tom spots yet another man, the third one. To him, the same question is asked: “Are you a noble or a savage?” The third man points in the direction of the first two men and he says: “Both are liars, they are savages” then he adds, “I am a noble”.

Question 1. Is the data given sufficient to tell us any man’s tribe? Give reasons for your answer.

Question 2. Is the data given sufficient to tell us each man’s tribe? Give reasons for your answer.

Question 3. To what extent can this puzzle ease our understanding of what critical thinking is and of what it consists?

123. Read the following passage then answer the questions that follow:

SERVANT BOY AND HIS BOSS

Jessie is a servant boy who every day drinks the wine of his Boss and puts water in the bottle to replace what he has drank. But the Boss having suspicions as to the quality of the wine, decides to buy Pastis (a French wine that changes colour if you add water). Jessie as usual, took a mouthful and added water to replace what he drank.

However, soon after he added water the Pastis became milky. When the Boss came back and noticed it, he was sure he had managed to nail Jessie as thief!!!

At that same moment, Jessie realized he was in trouble and decided to go into the kitchen.

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The Boss told his wife “Cherished, you will see, he will be obliged to acknowledge stealing”. He then shouted: “Jessie!”

Jessie answered: “Yes, Boss”.

Who drank my Pastis?” the boss asked, but there was No answer. The Boss reiterated his question: Still there was no answer from Jessie.

Then the Boss went to find Jessie in the kitchen and said to him: “Are You insane or what? Why when I call you, you say “Yes Boss” but when I ask you a question you don’t answer me?”

Jessie retorted that “It is that, boss, when you are in the kitchen there, you don’t understand anything at all, except the name” Then to prove that Jessie lies, the Boss says to him: “You stay besides Madam in the living room, me I go to the kitchen, and you ask me a question”.

Jessie accepted. The Boss went into the kitchen and Jessie shouted: “Boss”. He answered: “Yes, Jessie”.

Then Jessie asked: “Who goes to the House-girl’s bedroom when Madam is not there?” there was No answer from the kitchen. Jessie shouted again: “Boss, I say who made the House-girl pregnant?” No answer. He asked the Third time; “Boss, I say who made the House-girl pregnant?”

The Boss returns from the kitchen running and says “Jessie, It is true, you are right, when one is in the kitchen, one does not hear anything, only the name”.

- a) How should the Boss proceed in punishing Jessie for stealing the wine?
- b) Between Jessie and Boss, who is a better thinker and why? Give reasons
- c) What fallacy or fallacies are committed in this passage? Explain
- d) As a Critical and Creative thinker what lesson(s) can you learn from this passage?